GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Diploma Engineering

Level: Diploma

Branch: Plastics Engineering

Subject Code: DI04023031

Subject Name: Secondary Processing Techniques

w. e. f. Academic Year:	2025-26
Semester:	4th
Category of the Course:	PCC

Prerequisite:	Student should have knowledge regarding properties of various plastic material as well as other engineering materials and secondary plastic product manufacturing processes.
Rationale:	The changing demands of end users with respect to products led the plastic engineers for developing various kinds of processes. A plastic diploma engineer has to monitor operations of many specialized processes. This competency requires the knowledge of these specialized processes. Hence the course has been designed to develop this competency and its associated cognitive, practical and affective domain learning outcomes.

Course Outcome:

After Completion of the Course, Student will able to:

No	Course Outcomes	RBT Level
01	Troubleshoot Calendaring plant.	R,U
02	Perform Casting and Encapsulation process.	R,U,A
03	Select appropriate technique for FRP products.	R,U.A
04	Select suitable vinyl dispersion process for coating and molding.	U,A
05	Select suitable process for a cellular product.	U,A

**Revised Bloom's Taxonomy (RBT)*

Teaching and Examination Scheme:

Teaching Scheme (in Hours)			Total Credits L+T+ (PR/2)	Assessment Pattern and Marks				Total Marks
				Theory		Tutorial / Practical		
L	T	PR	C	ESE (E)	PA(M)	PA(I)	ESE (V)	
3	0	2	4	70	30	20	30	150

Course Content:

Unit No.	Content	No. of Hours	% of Weightage
Unit – I Calendaring Process	1.1 Introduction 1.2 Raw Material and Additives 1.3 Material Preparation (Premix, Blending, Gelation) 1.4 Roll Construction 1.5 Types Of Rolls – Cored And Drilled Rolls 1.6 Types Of Calendars ➢ Superimposed Calendars ➢ Offset Calendars ➢ Z Calendars 1.7 Heating And Lubrication Systems For Calendar Rolls 1.8 Start-Up And Shut-Down Procedure 1.9 Trouble Shooting 1.10 Compare With Sheet Extrusion. 1.11 Applications 1.12 Advantages And Disadvantages	10	22
Unit – II Casting And Encapsulation	2.1 Casting • Introduction • Casting Material • Casting Mold Materials • Casting Process • Applications • Advantages And Disadvantages 2.2 Encapsulation • Encapsulations Materials • Encapsulation Process • Applications	8	18
UNIT III Fiber Reinforced Plastics	3.1 Introduction 3.2 Introduction of various resins use in Composite 3.3 Fibers – Classification, Properties and Applications 3.4 Release Agents – Internal And External 3.5 Gel Coat Preparation and Its Application 3.6 Molding Compounds a. BMC b. SMC c. TMC	10	22

	3.8 Mould Materials 3.9 FRP Processes • Hand Lay Up and Spray Lay Up • Vacuum & Pressure Bag • Filament Winding • Pultrusion 3.10 Applications 3.11 Advantages And Disadvantages		
UNIT IV: Vinyl Dispersion	4.1 Introduction 4.2 Resin Selection Criteria 4.3 Plastisol Preparation 4.4 Vinyl Dispersion Process • Knife Coating • Roll Coating 4.5 Molding Process • Dip Coating And Dip Molding • Hot And Cold Moulding • Slush Molding 4.6 Application	8	18
UNIT V: Foam Processes	5.1 Introduction 5.2 General Production Methods 5.3 Blowing Agents – Physical and Chemical 5.4 Cellular Structure and Properties 5.5 Flexible And Rigid Foam of • Polyurethane (PU) • Poly Vinyl Chloride (PVC) • Polystyrene (PS) • Polyethylene (PE) • Epoxy and Silicone 5.6 Applications 5.7 Advantages And Disadvantages	9	20
	Total	**45**	**100**

Suggested Specification Table with Marks (Theory):

Distribution of Theory Marks (in %)					
R Level	**U Level**	**A Level**	**N Level**	**E Level**	**C Level**
20	**45**	**35**	---------	_____	_____

Where R: Remember; U: Understanding; A: Application, N: Analyze and E: Evaluate C: Create (as per Revised Bloom's Taxonomy)

References/Suggested Learning Resources:

(a) Books:

S.No.	Title of Book	Author	Publication with place, year and ISBN
1	Plastic Materials and Processes	S. S. Schwartz & S. H. Goodman	Van Nostrand Reinhold Company, 1992, 9780442227777
2	Plastic Engineering Handbook	J. L .Fredoz	Van Nostrand Reinhold Company, 1991, 978-1461576068
3	SPI Plastic Engineering Handbook	M. Bearins	Springer, 2012, 9781461576068
4	Calendaring of Plastics	Elden & Swan	American Elsevier Publishing Company, 1971, 9780592054391
5	FRP Technology	R. G. Weather head	Applied Science Publisher, 1980, 978-9400987234
6	Decoration and Assembly of Plastic Parts	Muccio E. A	ASM International, 1999, 9780871706348
7	Decorating Plastics	Margolis J.M	Hanser Publications, 1986, 9780029475805

(b) Open-source software and website:

1. https://www.youtube.com/watch?v=KSwqKPgaLqs
2. https://www.youtube.com/watch?v=RyEiIJB6asM
3. https://www.youtube.com/watch?v=_m29-u37TI8
4. https://www.youtube.com/watch?v=nwHgRGdjqOQ
5. https://www.youtube.com/watch?v=y2HlCMOe5p4
6. https://www.youtube.com/watch?v=wRIEl-5_uBE
7. https://www.youtube.com/watch?v=7iITFBVA_0w
8. Understanding the Basics of Foam Manufacturing - foam enthusiasts
9. nptel.ac.in/courses/112103308
10. Calendaring Process Basics: Knowing Its Principles And Application - LEADRP - Rapid Prototyping And Manufacturing Service
11. Notes (Synthetic Fibres and Plastics) PDF | PDF | Plastic | Synthetic Fiber

33. Suggested Practical List:

Sr. No.	Practical/Exercise (Course Outcomes in Psychomotor Domain according to NBA Terminology)	Approx. Hrs Required
1	To Prepare Raw Material for Calendaring process.	2
2	To set process variable of Calendaring Process for web.	2
3	To Select Suitable Calendar Roll arrangement for given product.	2
4	To prepare a product utilizing Casting process.	2
5	To prepare a product utilizing Encapsulation (Potting) Process.	4
6	To Prepare FRP Product using Hand lay-up process.	4
7	To Prepare FRP Product using Filament Winding process.	4
8	To Prepare hand gloves using Dip coating process.	4
9	To prepare product using slush moulding process.	2
10	To prepared PU foam using RIM process.	4

Suggested Activities for Students:

1. Assignments
2. Technical Quiz/MCQ Test
3. Students will prepare chart of various Calendaring process
4. Students will prepare application list of various Calendaring processes.
5. Students will prepare list of plastic product which require encapsulation application
6. Students will collect products with painting and coating application.
7. Students will prepare chart of various RFP processes.
8. Students will collect products decorated with various Foam processes.

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